

SUBTITLE SHIFTER

User & Deployment Guide

Version 1.0.0

A lightweight self-hosted web app for permanently shifting .srt subtitle timing from any browser.

Designed for Jellyfin, Plex, Emby and other home media servers.

At a glance: Search subtitle files, apply quick or custom time offsets, preview changes, and restore the original from an automatic .bak backup - all from a phone, tablet, or desktop browser.

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1. Overview

Subtitle Shifter is a lightweight self-hosted web application for fixing subtitles that are consistently too early or too late. It scans a configured media directory for .srt files, exposes them through a searchable browser interface, and permanently shifts every SRT timestamp by the amount you choose.

Best use case: You are watching a movie or episode in Jellyfin/Plex and the external SRT subtitle is consistently out of sync by a fixed amount.

The application is intentionally simple:

- No database is required.
- No manual file-path entry is required after setup.
- No video or audio files are modified.
- The first edit automatically creates a backup of the original subtitle.
- The web interface works from modern desktop and mobile browsers.

2. Features

- Automatically finds .srt subtitle files recursively under the configured media root.
- Searches by movie, show, season, episode, folder, or subtitle filename.
- Quick timing controls: -5s, -2s, -1s, -0.5s, +0.5s, +1s, +2s, +5s.
- Custom decimal offset such as -1.7 or +2.3 seconds.
- Built-in subtitle preview.
- Automatic .bak backup before the first modification.
- One-click restore of the original subtitle.
- Docker-ready deployment and Docker Compose example.
- Health-check endpoint for basic monitoring.
- Path traversal protection that keeps file operations inside MEDIA_ROOT.
- Mobile-friendly interface for use from a phone or tablet.

3. How It Works

1. Open Subtitle Shifter in a browser.
2. Search for the movie, TV show, season, episode, or subtitle file.
3. Select the required .srt file.
4. Choose a quick offset or enter a custom value.
5. Subtitle Shifter backs up the original file if a backup does not already exist.
6. Every timestamp in the SRT is shifted by the requested amount.
7. Reload playback in Jellyfin, Plex, Emby, or another player.

Input	Meaning
-2.0 s	Move every subtitle cue 2 seconds earlier
-0.5 s	Move every subtitle cue 0.5 seconds earlier
+0.5 s	Move every subtitle cue 0.5 seconds later

+2.0 s

Move every subtitle cue 2 seconds later

Example: If dialogue is heard first and the subtitle appears 2 seconds later, use -2 seconds. If the subtitle appears first and dialogue follows 2 seconds later, use +2 seconds.

4. Media Layout

The application recursively scans the configured media root. A typical home media structure looks like this:

```
/mnt/media/
├── FILMOVI/
│   ├── The Matrix/
│   │   ├── The Matrix.mkv
│   │   └── The Matrix.srt
│   └── SERIJE/
│       ├── Atlanta/
│       │   ├── Season 1/
│       │   │   ├── Atlanta - S01E01.mkv
│       │   │   ├── Atlanta - S01E01.srt
│       │   │   ├── Atlanta - S01E02.mkv
│       │   │   └── Atlanta - S01E02.srt
```

When the host directory is mounted as **/media**, Subtitle Shifter searches all child folders automatically.

5. Docker Installation

5.1 Clone the repository

```
git clone https://github.com/YOUR_USERNAME/subtitle-shifter.git
cd subtitle-shifter
```

Replace **YOUR_USERNAME** with the GitHub account that hosts the project.

5.2 Build the image

```
docker build -t subtitle-shifter .
```

5.3 Start the container

```
docker run -d \
  --name subtitle-shifter \
  --restart unless-stopped \
  -e MEDIA_ROOT=/media \
  -e PORT=5070 \
  -p 5070:5070 \
  -v /mnt/media:/media:rw \
  subtitle-shifter
```

Important: The host media path must be mounted read/write (:rw). The app needs write access to modify .srt files and create .bak backups.

5.4 Open the interface

```
http://SERVER_IP:5070
```

Example on a local network:

```
http://192.168.1.100:5070
```

6. Docker Compose

If Docker Compose is installed, the included compose file can build and start the application in one command:

```
docker compose up -d --build
```

Default volume mapping:

```
volumes:
  - /mnt/media:/media:rw
```

If your media lives somewhere else, change only the host side of the mapping:

```
volumes:
  - /your/media/path:/media:rw
```

7. Configuration

Variable	Default	Purpose
MEDIA_ROOT	/media	Root directory recursively scanned for .srt files
PORT	5070	Web server port inside the container
PREVIEW_LINES	80	Number of subtitle lines shown in the preview panel

A typical container configuration is:

```
-e MEDIA_ROOT=/media
-e PORT=5070
-p 5070:5070
-v /mnt/media:/media:rw
```

8. Mobile and Remote Use

The interface is designed to be usable from a phone, tablet, or desktop browser. A typical mobile workflow is:

```
Phone
↓
Subtitle Shifter
↓
Select episode.srt
↓
Apply +2 seconds
↓
Reload Jellyfin playback
```

For remote access, use a trusted private network such as:

- Tailscale
- WireGuard
- another trusted VPN
- an authenticated reverse proxy

Security: Do not expose port 5070 directly to the public internet without authentication. Version 1.0.0 does not include its own login system.

9. Backups and Restore

Before a subtitle is modified for the first time, Subtitle Shifter creates a sibling backup:

```
episode.srt
episode.srt.bak
```

The backup is not overwritten by later shifts. This means the Restore original button returns to the subtitle state from before the first edit, not merely the previous edit.

Safe behavior: Video files, audio files, and media-server databases are not modified by the application.

10. Permissions and Troubleshooting

10.1 Subtitles are visible but shifting fails

This almost always means the container can read the media mount but cannot write to it.

Check the container logs:

```
docker logs subtitle-shifter
```

A typical permissions error looks like:

```
PermissionError: [Errno 13] Permission denied
```

Verify that the volume is mounted read/write:

```
-v /mnt/media:/media:rw
```

Also check the host file permissions for the affected subtitle. The application must be able to overwrite the .srt file and create the .bak file in the same directory.

10.2 Container is not running

```
docker ps --filter name=subtitle-shifter
```

If needed, inspect the last log lines:

```
docker logs --tail 100 subtitle-shifter
```

10.3 Port conflict

If port 5070 is already used by another service, map a different host port:

```
-p 5090:5070
```

Then open:

```
http://SERVER_IP:5090
```

11. Health Check and API

11.1 Health endpoint

```
GET /health
```

Example response:

```
{
  "ok": true,
  "media_root": "/media",
  "media_root_exists": true,
  "version": "1.0.0"
}
```

11.2 List subtitle files

```
GET /api/files
```

11.3 Preview a subtitle

```
GET /api/preview?path=SERIJE/Show/Season%201/example.srt
```

11.4 Shift a subtitle

```
POST /api/shift
Content-Type: application/json

{
  "path": "SERIJE/Show/Season 1/example.srt",
  "delta_ms": 2000
}
```

11.5 Restore original

```
POST /api/restore
Content-Type: application/json

{
  "path": "SERIJE/Show/Season 1/example.srt"
}
```

12. Security

Subtitle Shifter is an administrative media-management tool because it has write access to subtitle files. Version 1.0.0 intentionally has no built-in authentication.

- Run it on a trusted LAN or private VPN.
- Do not forward port 5070 directly from the public internet.
- If public exposure is required, place it behind an authenticated reverse proxy.
- Limit the mounted host directory to the media tree that actually needs subtitle editing.
- Keep backups enabled; they are the primary rollback mechanism.

13. Current Limitation: Fixed Offset Only

Subtitle Shifter v1.0.0 fixes a constant timing offset. For example:

```
Beginning: subtitle is 2 seconds late
Middle:   subtitle is 2 seconds late
End:     subtitle is 2 seconds late
```

This is exactly the case the current application is designed to solve.

It does not yet correct progressive drift such as:

```
Beginning: synchronized
20 min:   1 second late
40 min:   2 seconds late
```

That kind of error usually needs time scaling, two-point synchronization, or FPS correction rather than a simple offset.

14. Planned Features

- Subtitle stretch / FPS correction.
- Two-reference-point synchronization.
- Built-in authentication.
- Multiple users and permissions.
- Direct subtitle text editing.
- Jellyfin library refresh integration.
- Additional UI themes.
- Recently modified subtitle history.
- Language filtering.
- Backup management.

15. Development

Local development can be done without Docker:

```
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
MEDIA_ROOT=/path/to/media python app.py
```

Then open:

```
http://localhost:5070
```

Technology stack

- Python
- Flask
- HTML
- CSS
- JavaScript

- Docker

No database is required.

16. Contributing and License

Bug reports, feature requests, and pull requests are welcome when the project is hosted publicly on GitHub.

Suggested repository topics:

```
jellyfin plex subtitles srt subtitle-sync docker  
self-hosted media-server flask python
```

The project is intended to be distributed under the MIT License. See the repository LICENSE file for the full license text.

Project summary: Subtitle Shifter turns a common media-server annoyance into a quick browser workflow: find the SRT, adjust timing, verify playback, and restore the original whenever needed.